TO21COMMS	발행일 2024-10-25	버전 1.0	문서번호 DataEventTrackingService 배포 프로세스		
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제목 DataEventTrackingService 배포 프로세스		Title Data Event Tracking Service Deployment Process			

(총괄 1 세부) 지능형 디지털 트윈 연합 운용 및 예측 핵심 기술 개발

DataEventTrackingService

배포 프로세스

(v1.0)

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문서 정보

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개정이력

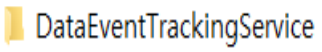
버전	개정 일자	내용	작성자	확인자
1.0	2024-10-25	항목 검토 및 정보 수정	김경진	

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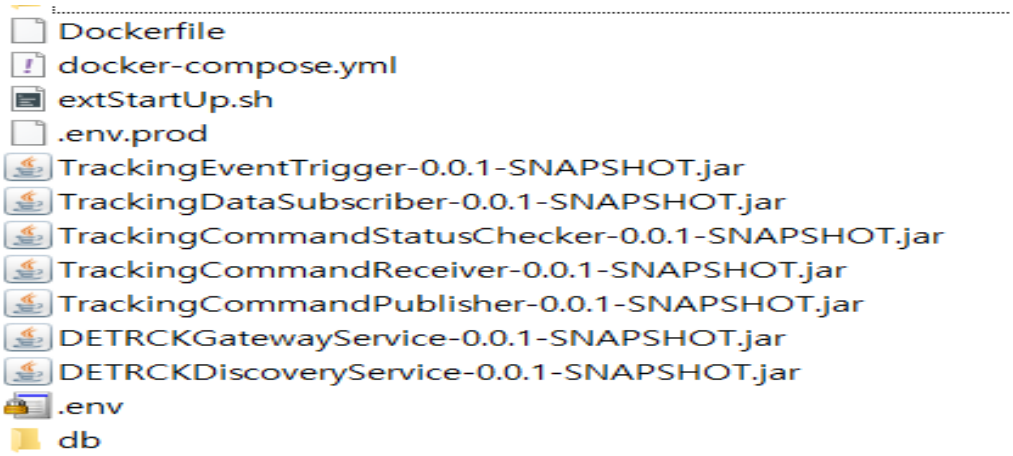
1. 동적 데이터/이벤트 트래킹

A. FTP 서버 접속 ex) FileZila

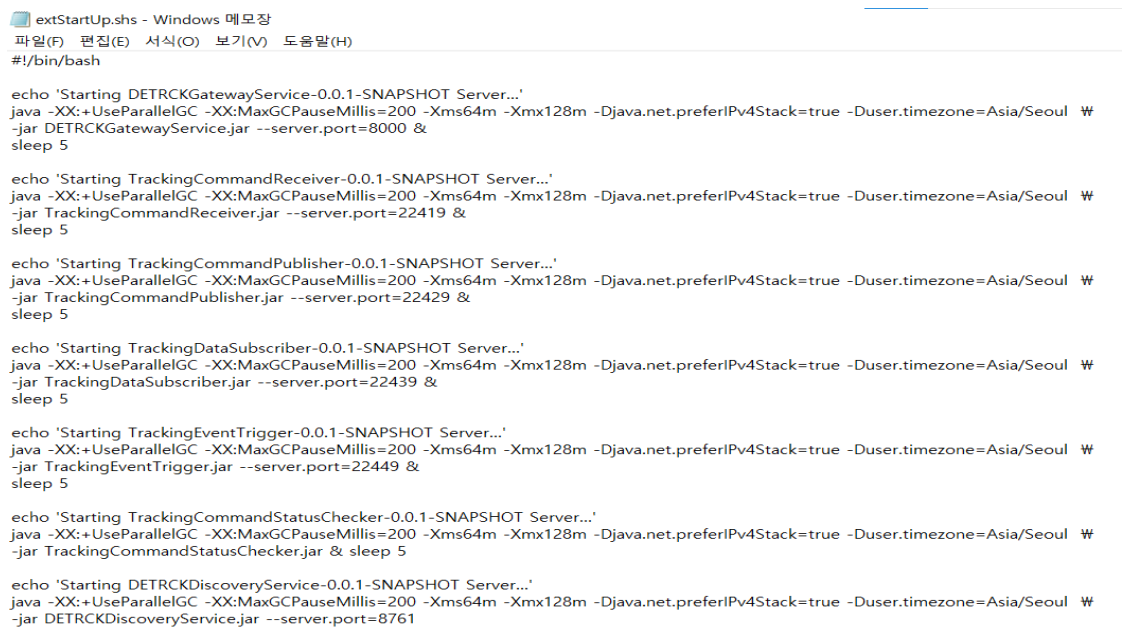
B. /usr/local 경로 하위에 하기와 같이 폴더 생성



C. B에서 만든 폴더 하위에 7개의 jar 파일, env.prod(스프링부트 설정파일), extStartUp.sh, Dockerfile, docker-compose.yml, .env(DB 계정정보 설정파일), db(데이터 베이스 설정파일) 파일 서버 업로드



D. Jar 파일 실행을 위한 쉘파일(extStartUp.sh)



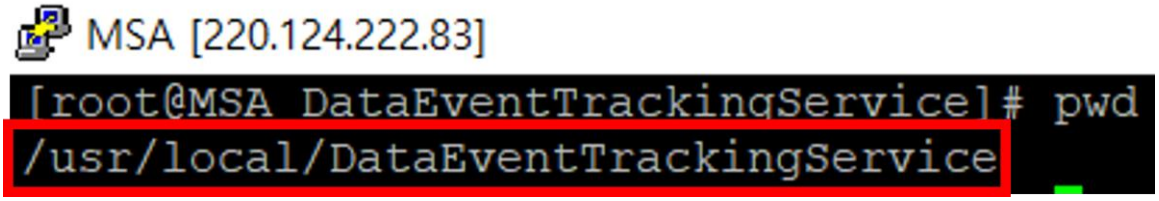
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E. Docker image 생성을 위한 Dockerfile

```
FROM adoptopenjdk/openjdk11:latest
ARG JAR_FILE1=DETRCKDiscoveryService-0.0.1-SNAPSHOT.jar
ARG JAR_FILE2=DETRCKGatewayService-0.0.1-SNAPSHOT.jar
ARG JAR_FILE3=TrackingCommandReceiver-0.0.1-SNAPSHOT.jar
ARG JAR_FILE4=TrackingCommandPublisher-0.0.1-SNAPSHOT.jar
ARG JAR_FILE5=TrackingDataSubscriber-0.0.1-SNAPSHOT.jar
ARG JAR_FILE6=TrackingEventTrigger-0.0.1-SNAPSHOT.jar
ARG JAR_FILE7=TrackingCommandStatusChecker-0.0.1-SNAPSHOT.jar
COPY ${JAR_FILE1} DETRCKDiscoveryService.jar
COPY ${JAR_FILE2} DETRCKGatewayService.jar
COPY ${JAR_FILE3} TrackingCommandReceiver.jar
COPY ${JAR_FILE4} TrackingCommandPublisher.jar
COPY ${JAR_FILE5} TrackingDataSubscriber.jar
COPY ${JAR_FILE6} TrackingEventTrigger.jar
COPY ${JAR_FILE7} TrackingCommandStatusChecker.jar
ADD extStartUp.sh /
RUN chmod +x /extStartUp.sh
CMD ["/extStartUp.sh"]
```

F. 터미널 접속 Ex) Putty

G. 현재 경로 확인 - pwd 명령어 입력



H. 폴더 내 파일 리스트 확인 - 명령어 ll 입력

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MSA [220.124.222.83]

```
[root@MSA DataEventTrackingService]# ll
합계 373836
-rw-r--r-- 1 jejutp jejutp 57002302 10월 16 14:54 DETRCKDiscoveryService-0.0.1-SNAPSHOT.jar
-rw-r--r-- 1 jejutp jejutp 50024626 10월 16 14:54 DETRCKGatewayService-0.0.1-SNAPSHOT.jar
-rwxr-xr-x 1 jejutp jejutp      822  7월  5 14:36 Dockerfile
-rw-r--r-- 1 jejutp jejutp 45427876 10월 22 20:42 TrackingCommandPublisher-0.0.1-SNAPSHOT.jar
-rw-r--r-- 1 jejutp jejutp 71501123 10월 22 20:43 TrackingCommandReceiver-0.0.1-SNAPSHOT.jar
-rw-r--r-- 1 jejutp jejutp 45411585 10월 16 14:55 TrackingCommandStatusChecker-0.0.1-SNAPSHOT.jar
-rw-r--r-- 1 jejutp jejutp 83164874 10월 24 18:47 TrackingDataSubscriber-0.0.1-SNAPSHOT.jar
-rw-r--r-- 1 jejutp jejutp 30251048 10월 24 19:54 TrackingEventTrigger-0.0.1-SNAPSHOT.jar
drwxr-xr-x 5 jejutp jejutp       48 10월 16 14:55 db
-rwxr-xr-x 1 jejutp jejutp     2057  7월 18 14:28 docker-compose.yml
-rwxr-xr-x 1 jejutp jejutp     1786  7월 18 14:25 extStartUp.sh
```

I. E에서 생성한 Dockerfile command로 build

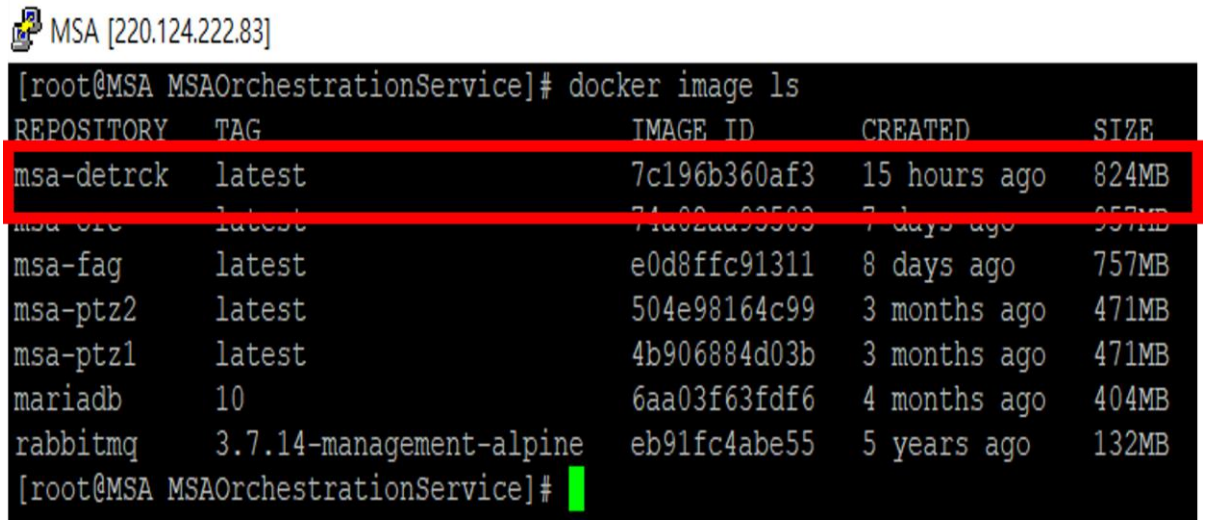
i. docker build -t msa-detrck .(이미지 생성 명령어)

MSA [220.124.222.83]

```
[root@MSA DataEventTrackingService]# docker build -t msa-detrck .
[+] Building 1.6s (15/15) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 861B
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load metadata for docker.io/adoptopenjdk/openjdk11:latest
 => [ 1/10] FROM docker.io/adoptopenjdk/openjdk11:latest@sha256:e833749689e8be972872d383a89a77fa023e754fa
 => [internal] load build context
 => => transferring context: 481B
 => CACHED [ 2/10] COPY DETRCKDiscoveryService-0.0.1-SNAPSHOT.jar DETRCKDiscoveryService.jar
 => CACHED [ 3/10] COPY DETRCKGatewayService-0.0.1-SNAPSHOT.jar DETRCKGatewayService.jar
 => CACHED [ 4/10] COPY TrackingCommandReceiver-0.0.1-SNAPSHOT.jar TrackingCommandReceiver.jar
 => CACHED [ 5/10] COPY TrackingCommandPublisher-0.0.1-SNAPSHOT.jar TrackingCommandPublisher.jar
 => CACHED [ 6/10] COPY TrackingDataSubscriber-0.0.1-SNAPSHOT.jar TrackingDataSubscriber.jar
 => CACHED [ 7/10] COPY TrackingEventTrigger-0.0.1-SNAPSHOT.jar TrackingEventTrigger.jar
 => CACHED [ 8/10] COPY TrackingCommandStatusChecker-0.0.1-SNAPSHOT.jar TrackingCommandStatusChecker.jar
 => CACHED [ 9/10] ADD extStartUp.sh /
 => CACHED [10/10] RUN chmod +x /extStartUp.sh
 => exporting to image
 => => exporting layers
 => => writing image sha256:7c196b360af3655c358f157dc78b9f850dd9d641e05bc571cfc902a64f94f6b4
 => => naming to docker.io/library/msa-detrck
```

ii. 이미지 리스트 확인 - 명령어 dokcer image ls

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J. env 및 env.prod 파일은 서버 환경에 따라서 수정

K. DB 설정

i. db폴더 생성하고 하위에 initdb.d , data, conf.d 3개 폴더 생성

ii. initdb.d 폴더 하위에 create_table.sql(스크립트 문) 파일 업로드

iii. conf.d 폴더 하위에 my.cnf(설정 파일) 파일 업로드

L. Docker container 생성을 위한 docker-compose.yml

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```
version: '3'
services:
  dataeventtrackingservice:
    container_name: dataeventtrackingservice
    image: msa-detrck
    ports:
      - 8763:8761
      - 8002:8000
      - 22419:22419
      - 22429:22429
      - 22439:22439
      - 22449:22449
      - 22459:22459
    volumes:
      - type: bind
        source: /data/DtEvtTracking/trackCmdReceiver
        target: /hdd/DtEvtTracking/trackCmdReceiver
      - type: bind
        source: /data/DtEvtTracking/trackCmdPublisher
        target: /hdd/DtEvtTracking/trackCmdPublisher
      - type: bind
        source: /data/DtEvtTracking/trackDataSub
        target: /hdd/DtEvtTracking/trackDataSub
      - type: bind
        source: /data/DtEvtTracking/trackEntTrigger
        target: /hdd/DtEvtTracking/trackEntTrigger
      - type: bind
        source: /data/DtEvtTracking/trackCmdStsChecker
        target: /hdd/DtEvtTracking/trackCmdStsChecker
    env_file: .env.prod
    networks:
      - detrck-network
    restart: unless-stopped
```

⇨ 컨테이너 및 빌드한 이미지명 작성

⇨ Fort Forwarding

⇨ Mount 정보 및 설정파일 세팅

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```
rabbitmq:
  container_name: rabbitmq_detrck
  image: rabbitmq:3.7.14-management-alpine
  volumes:
    - /hdd/rabbitmq/data/detrck/:/var/lib/rabbitmq/mnesia
    - /hdd/rabbitmq/logs/detrck:/var/log/rabbitmq/
  environment:
    - RABBITMQ_DEFAULT_USER=guest
    - RABBITMQ_DEFAULT_PASS=guest
  ports:
    - "5674:5672"
    - "15674:15672"
    - "25674:25672"
    - "4371:4369"
    - "5669:5671"
    - "15669:15671"
    - "1885:1883"
    - "8885:8883"
    - "61617:61613"
    - "61618:61614"
  networks:
    - detrck-network
  restart: unless-stopped
```

⇨ RabbitMq 정보 셋팅

```
mariadb:
  image: mariadb:10
  ports:
    - 21219:3306
  volumes:
    - ./db/conf.d:/etc/mysql/conf.d
    - /hdd/dataTest/mariadb3:/hdd/data/mariadb
    - ./db/initdb.d:/docker-entrypoint-initdb.d
  env_file: .env
  environment:
    TZ: Asia/Seoul
  networks:
    - detrck-network
  restart: unless-stopped
```

⇨ MariaDB 정보 셋팅

```
networks:
 detrck-network:
   ipam:
     driver: default
     config:
       - subnet: 172.20.0.0/16
         gateway: 172.20.0.1
```

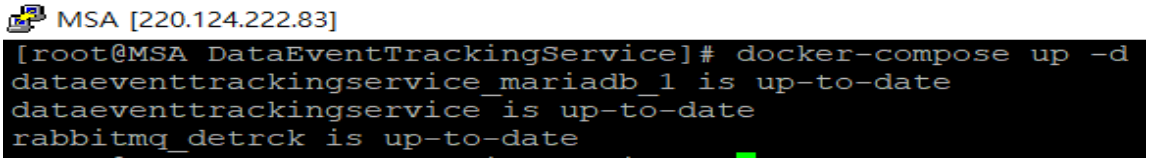
⇨ 컨테이너 ip 설정

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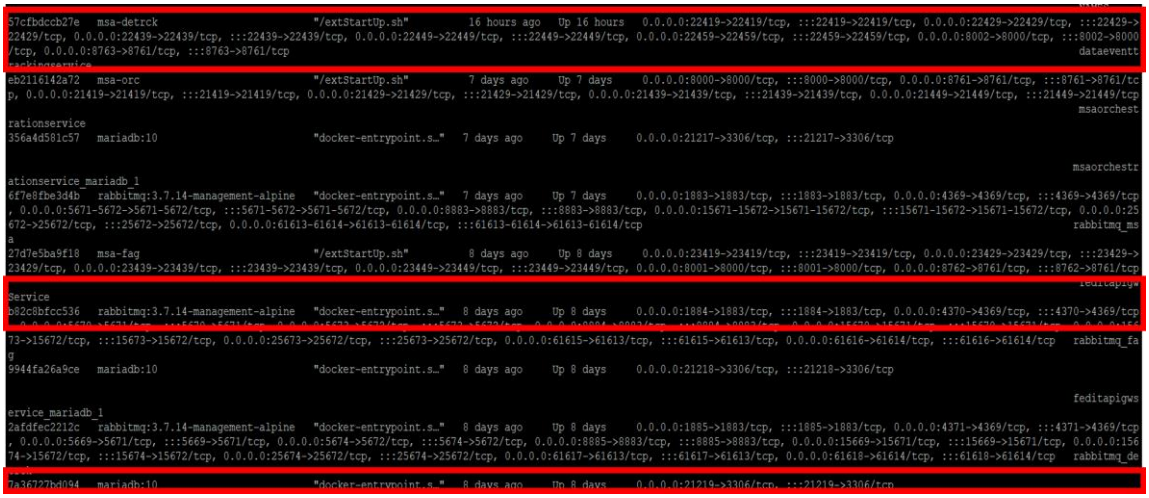
M. I에서 생성 docker-compose.yml command로 build

i. F,G 단계 확인

ii. dokcer-compose up -d(컨테이너 생성 명령어) 입력

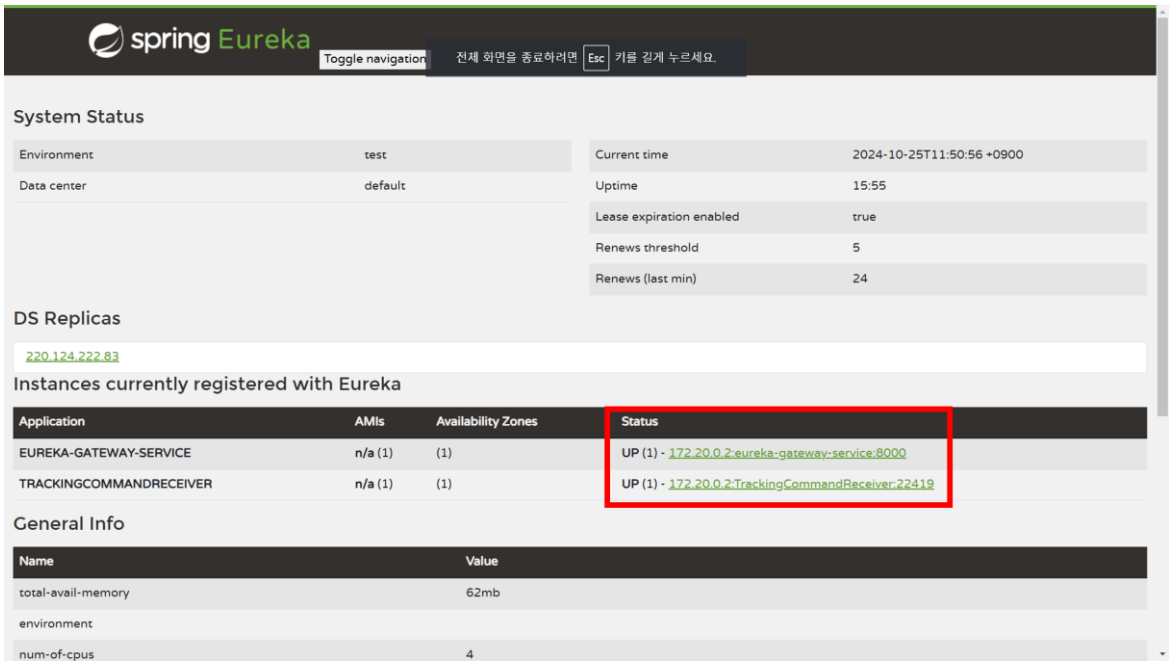


iii. 컨테이너 리스트 확인 - 명령어 docker ps -a

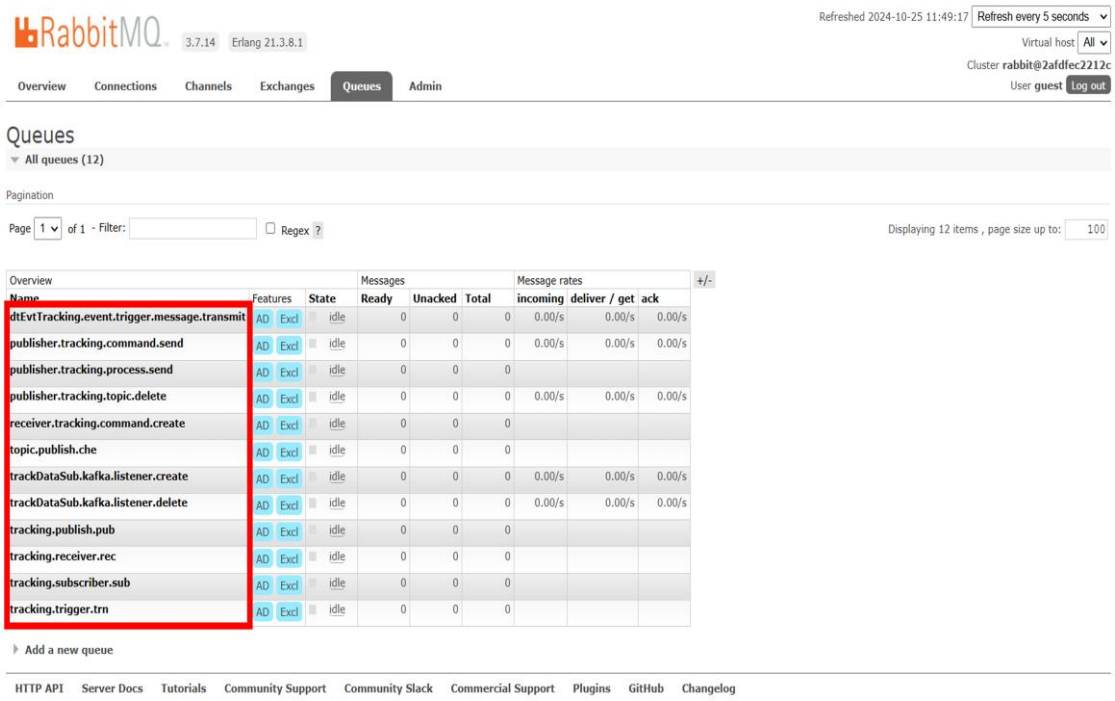


N. Euerka 서버 확인 => url :http://220.124.222.83:8763

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0. RabbitMq 서버 확인 => url : http://220.124.222.83:15674



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